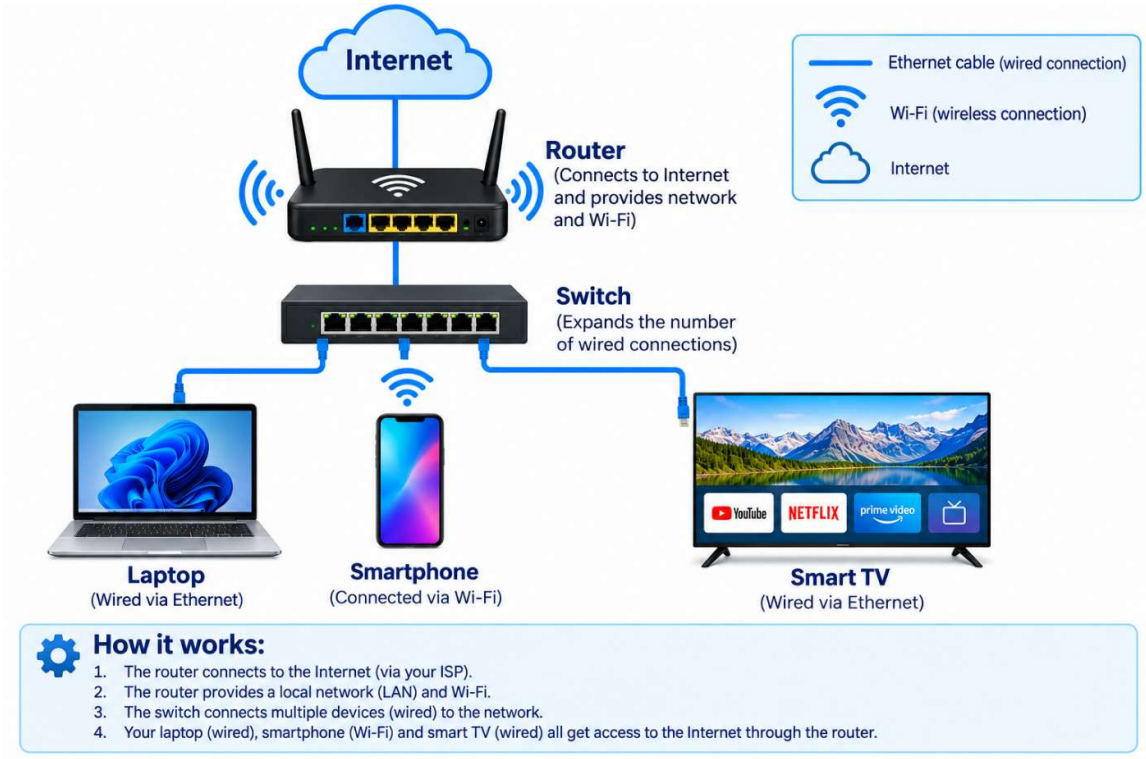
Assignment – 1.6

Network devices

Q - 1. Draw and label a simple home network diagram showing a router, switch, and three devices (like a laptop, smartphone, and smart TV), indicating how each device connects to the network.



<u>Q - 2.</u> Create a comparison table listing at least four network devices (router, switch, hub, access point) and describe one real-world use case for each, referencing apps or services you use daily (e.g., WiFi for streaming on YouTube, LAN gaming, etc.).

Network Device	Use	Real-World Example
Router	Connects networks to the internet.	Using YouTube on a mobile phone through Wi-Fi.
Switch	Connects multiple devices in a LAN.	Connecting computers in a computer lab.
Hub	Sends data to all connected devices.	Connecting computers in an old computer lab.
Access Point	Provides wireless Wi-Fi connectivity.	Connecting a laptop to Wi-Fi for Google Classroom.

<u>Q - 3.</u> Write a short scenario where you explain how data travels from your phone to the internet when you order food on Zomato, mentioning the role of each network device involved (router, switch, access point).

➢ **Scenario: Ordering Food on Zomato**

When I order food on Zomato using my phone:

1. **Access Point:** Connects my phone to Wi-Fi.
2. **Router:** Sends my order to the internet.
3. **Switch:** Transfers data between connected devices.
4. **Internet:** Sends my order to Zomato and brings the confirmation back to my phone.

<u>Q - 4.</u> Given this situation: you have a WiFi router, but your gaming PC only has an Ethernet port and is far from the router. Suggest two different network devices you could use to connect your PC to the internet and explain how each would work in this setup. Hint: Think about devices like switches, powerline adapters, or WiFi extenders.

- **Powerline Adapter:** It uses the home's electrical wiring to carry internet signals from the router to the gaming PC. Connect one adapter to the router and another near the PC, then use Ethernet cables.

- **WiFi Extender:** It receives the WiFi signal from the router and extends it closer to the PC. Connect the PC to the extender using an Ethernet cable.